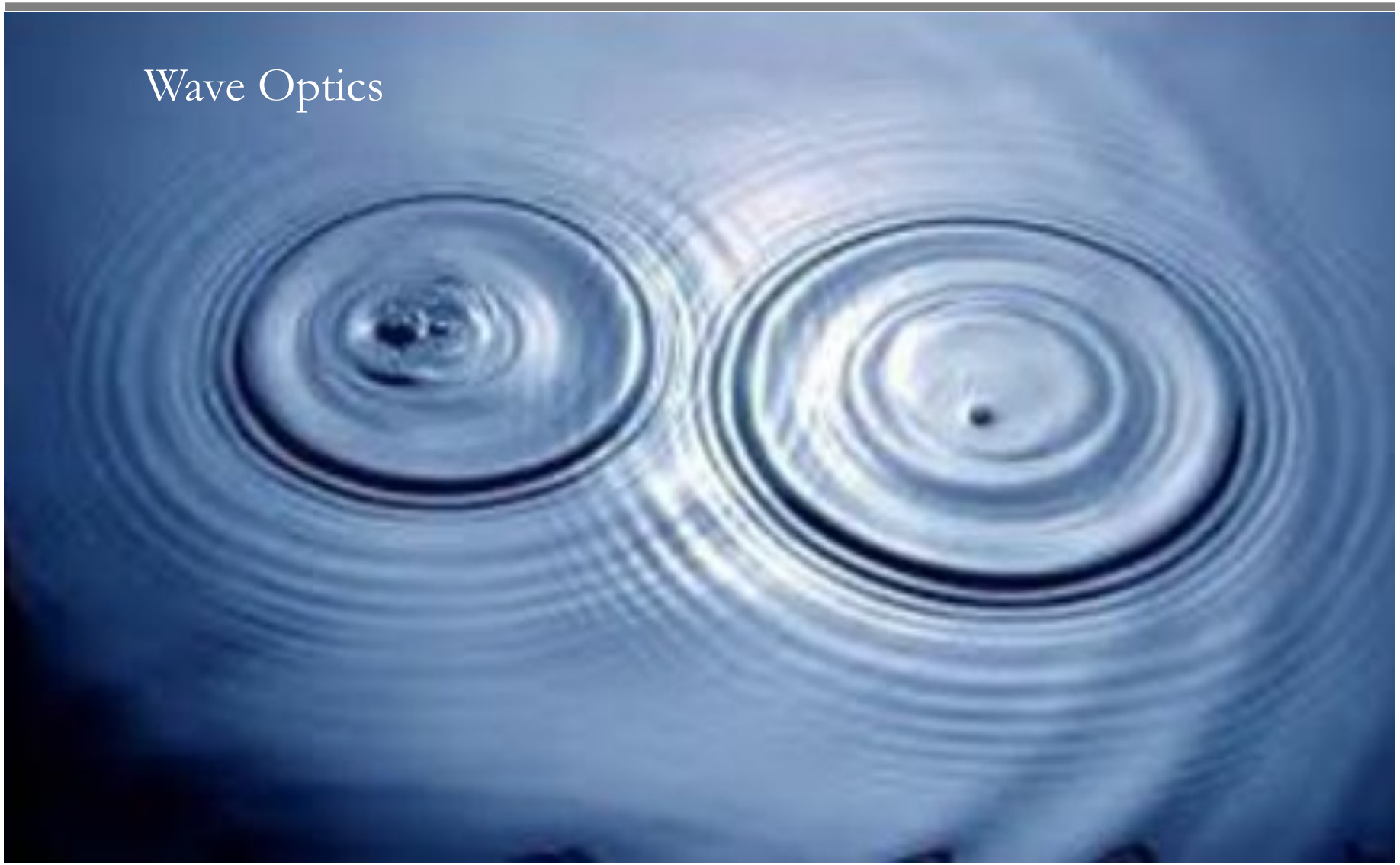

Wave Optics

Foundation

Wave optics can better describe the different phenomena involving light, mainly those associated with the wave nature of light such as interference. Objects comparable in size to optical wavelength can be treated more adequately than ray optics. However, the description of light-matter interaction is limited.

In wave optics, light is considered as a wave traveling in space with a defined amplitude, speed and wavelength, just like mechanical (sound) waves. Wave polarization is not relevant.

The guiding principles are:

1) Light is described by the scalar wave equation.
2) Material is represented by the refractive index (velocity ratio) and attenuation.
3) The light wave has a specified amplitude, velocity and wavelength.
4) Amount of reflection/transmission and wave polarization cannot be determined.

The scalar wave equation

$$\nabla^2 u - \frac{1}{v^2}\frac{d^2 u}{dt^2} = 0$$

The Scalar Wave Equation

The scalar wave equation

$$\nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$$

$U(r,t)$ is wave function

Assume that the wave function is time-harmonic

$$U(r,t) = U(r)\, e^{j\omega t}$$

The scalar wave equation becomes

$$\nabla^2 U(r) + \left(\frac{\omega}{v}\right)^2 U(r) = 0$$

$$(\nabla^2 + k^2)\, U(r) = 0$$

Put $k = \frac{\omega}{v}$ Wave number (rad/m)

$$k = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi}{\lambda}$$

$$v = \frac{v_o}{n}$$

$$k = nk_o$$

$$\lambda = \frac{\lambda_o}{n}$$

A general solution is of the form

$$U(r) = a(r) \cdot e^{j\phi(r)}$$

A general time-harmonic solution is of the form

$$U(r,t) = a(r) \cdot e^{j\phi(r)} \cdot e^{j\omega t}$$

An instantaneous solution is of the form

$$u(r,t) = a(r) \cdot \cos[\omega t + \phi(r)]$$

$$\phi(r) = kr$$

$$u(r,t) = \frac{1}{2}[U(r,t) + U^*(r,t)]$$

Some Wave Types

A number of possible forms of the solutions $U(r) = a(r) \cdot e^{j\phi(r)}$

Spherical waves $U(r) = \frac{A}{r} e^{-jkr}$

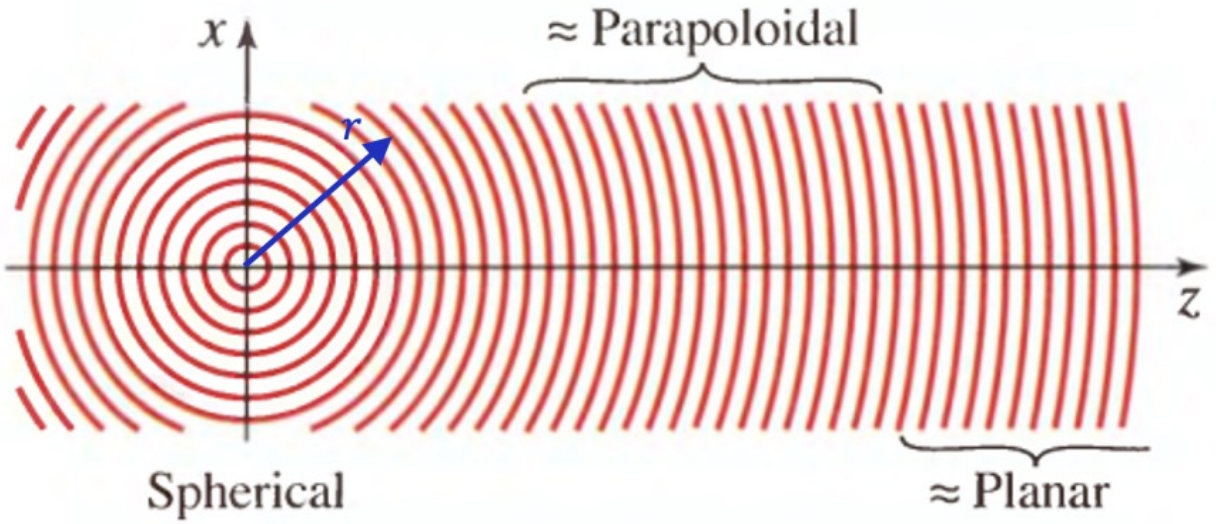
Plane waves $U(r) = A\, e^{-jkz}$

Parapoloidal waves $U(r) = \frac{A}{z} e^{-jkz}\, e^{-jk\left(\frac{x^2+y^2}{2z}\right)}$

A is in general a complex value.

r, x, y, z is the direction of wave propagation.

k is the propagation/phase constant (rad/m)



Wave Intensity and Power

1) Optical intensity of a wave is given by

$$I(r) = |U(r)|^2$$

Time-harmonic

Monochromatic light

Such that

$$U(r) = \sqrt{I(r)} \cdot e^{j\phi(r)}$$

Instantaneous intensity

$$I(r,t) = < U(r,t) \cdot U^*(r,t) >$$

2) Wave power is given by

$$P = \int_A I(r)\, dA$$

Spherical waves

$$U(r) = \frac{\sqrt{I}}{r} e^{-jkr}$$

Plane waves

$$U(r) = \sqrt{I}\, e^{-jkz}$$

Interference (1)

1) Two monochromatic plane light waves, same wavelength.

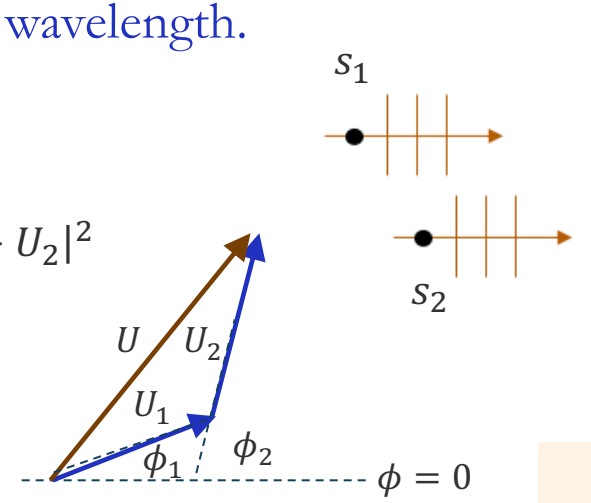
$$U_1 = \sqrt{I_1}\cdot e^{j\phi_1} \qquad U_2 = \sqrt{I_2}\cdot e^{j\phi_2}$$

$$U(r) = U_1 + U_2 \quad \Rightarrow \quad I(r) = |U(r)|^2 = |U_1 + U_2|^2$$

$$I(r) = |U_1|^2 + |U_2|^2 + U_1U_2^* + U_1^*U_2$$

$$I(r) = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos(\phi_2 - \phi_1)$$

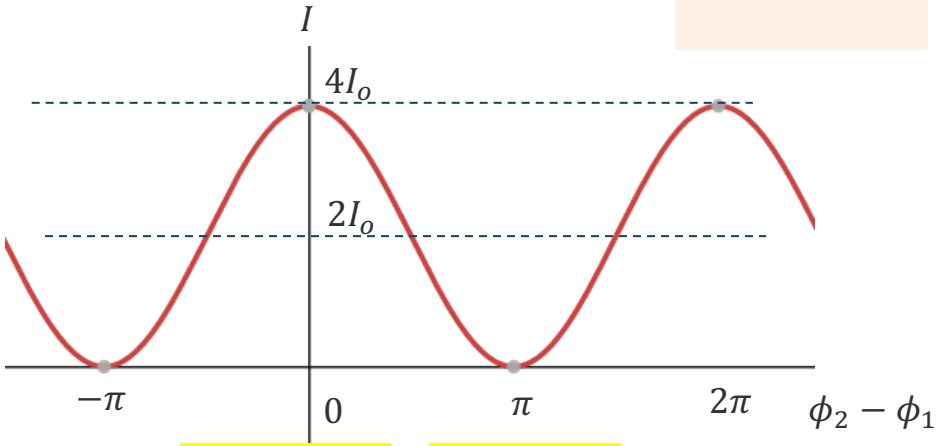
Interference equation



Example. Draw the interference pattern for two sources with $I_1 = I_2 = I_o$

$$I = 2I_o[1 + \cos(\phi_2 - \phi_1)]$$

What is the pattern on the screen?



Interference (1)

Where does the phase come from?

1) Sources are out-of-phase.

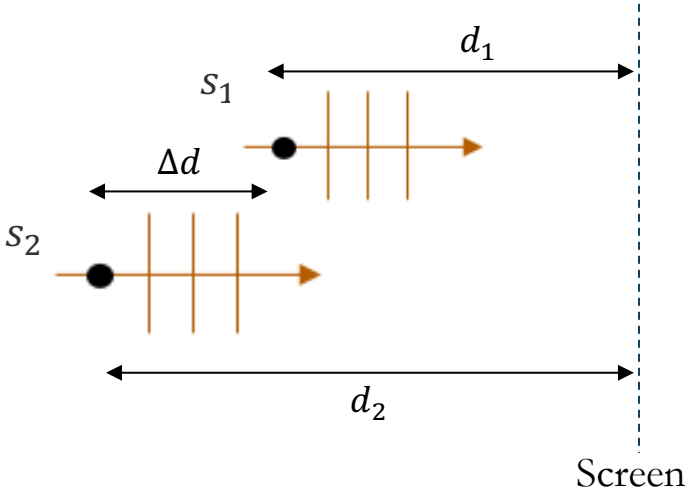
2) Distance.

$$U_1 = \sqrt{I_o} \cdot e^{-jkd_1} \qquad U_2 = \sqrt{I_o} \cdot e^{-jkd_2}$$

$$I = 2I_o[1 + \cos(k\Delta d)]$$

$$k\Delta d = \frac{2\pi}{\lambda}\Delta d \qquad \Delta d = \pm m\lambda \quad \text{Constructive}$$

$$\Delta d = \pm\frac{(2m+1)\lambda}{2} \quad \text{Destructive} \qquad m = 0, 1, 2, \ldots$$



3) Properties of path (n).

$$U_1 = \sqrt{I_o} \cdot e^{-jk_1 d} \qquad U_2 = \sqrt{I_o} \cdot e^{-jk_2 d}$$

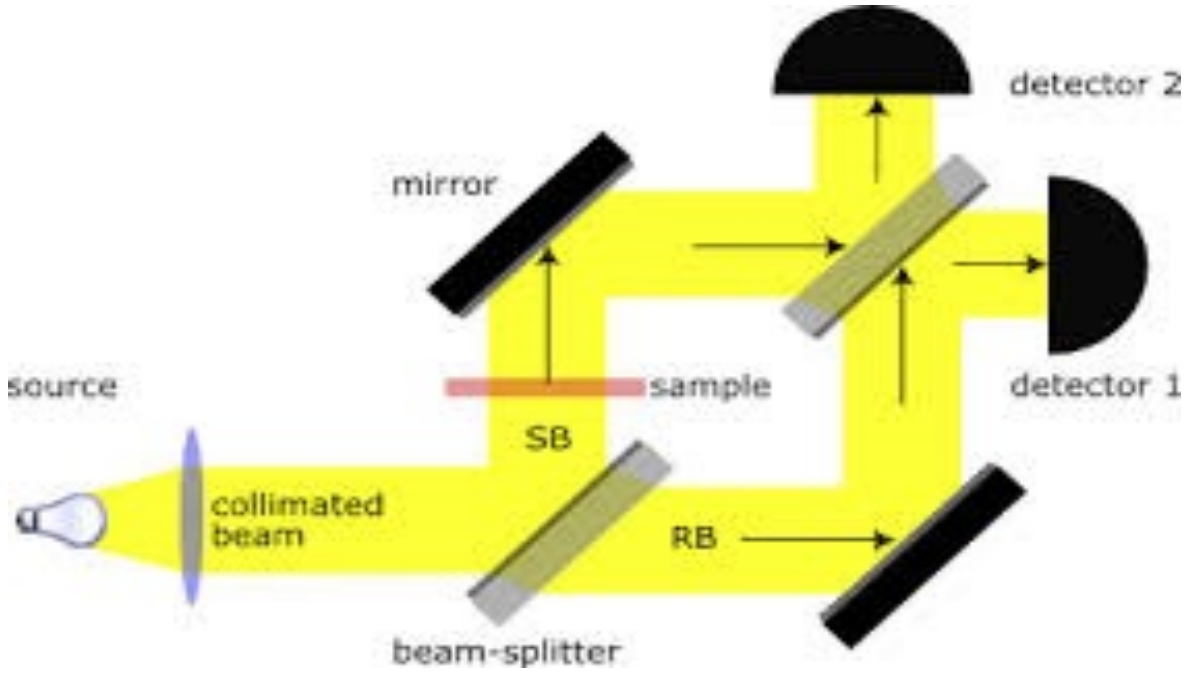
$$I = 2I_o[1 + \cos(k_2 - k_1)d]$$

$$(k_2 - k_1)d = \frac{2\pi}{\lambda} d(n_2 - n_1) = \frac{2\pi d}{\lambda}\Delta n$$

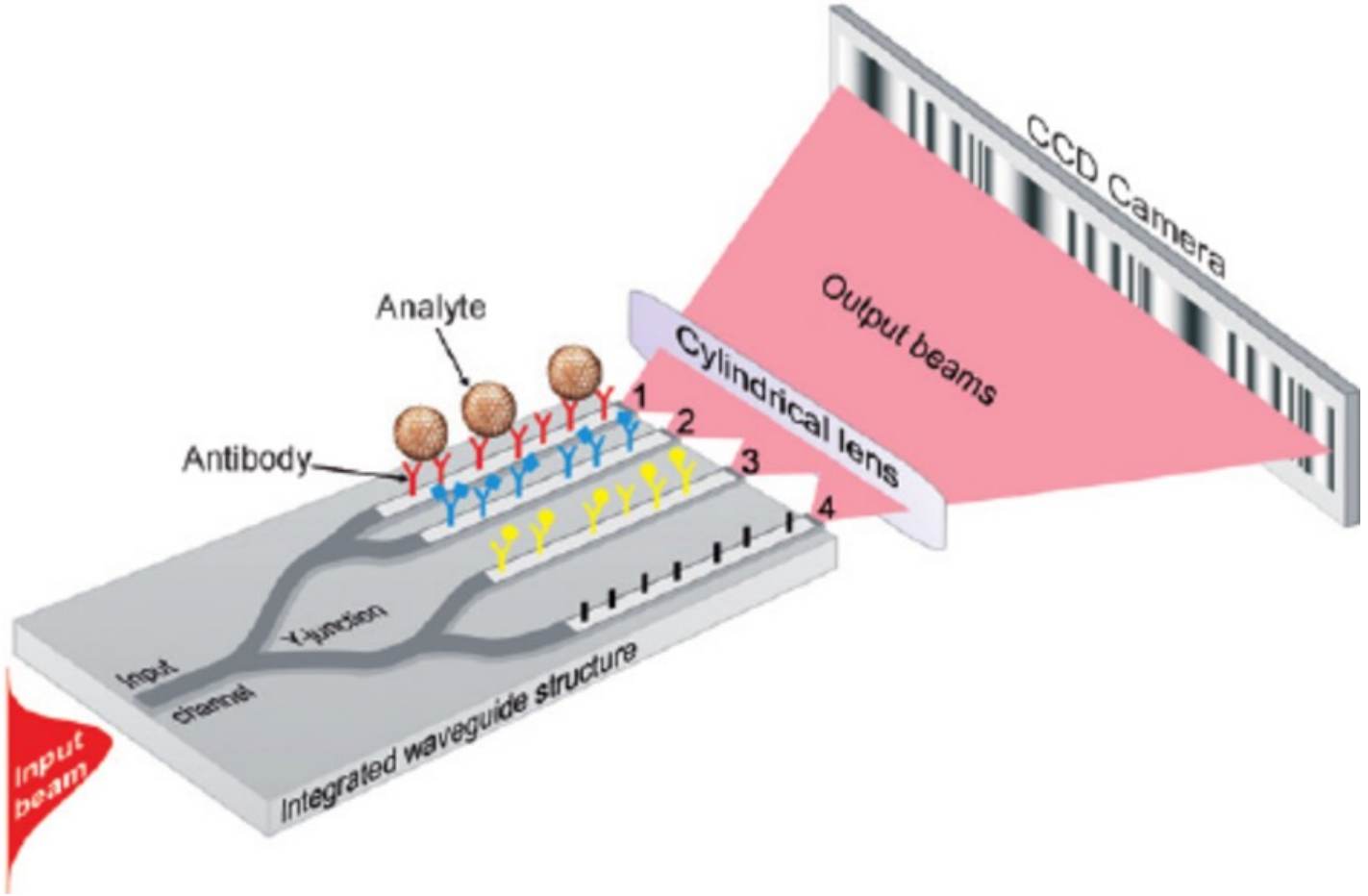
Applications: Interferometry

1) Strain/stress
2) Index change (uniformity)
3) Layer thickness.

Interferometer



Interferometry in Biosensing



Interference (2)

2) Two monochromatic plane light waves, same wavelength, at oblique angle.

$$U_1 = \sqrt{I_o} \cdot e^{-jkz} \qquad U_2 = \sqrt{I_o} \cdot e^{-jk(x \sin\theta + z\cos\theta)}$$

At the screen,

$$U_1 = \sqrt{I_o} \qquad U_2 = \sqrt{I_o} \cdot e^{-jkx \sin\theta}$$

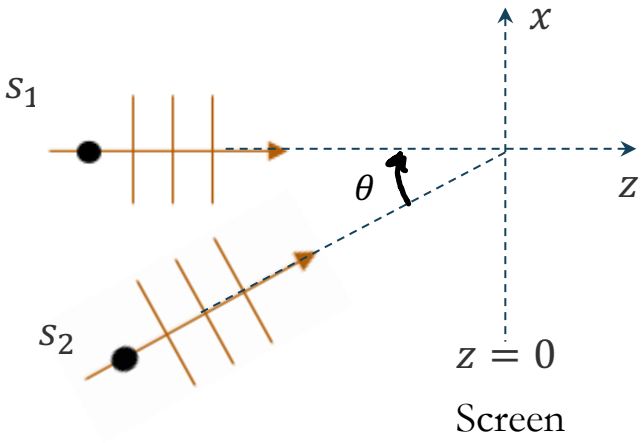
$$I = 2I_o[1 + \cos(kx \sin\theta)]$$

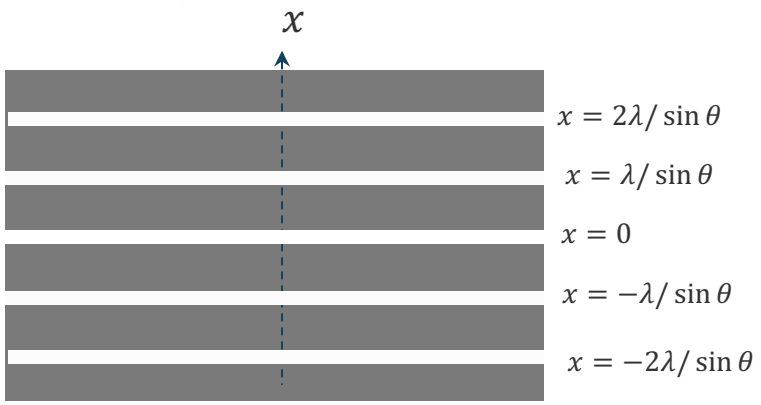
$$kx \sin\theta = m(2\pi) \quad \text{Maxima} \quad m = 0, 1, 2, \ldots$$

$$x_{max} = \frac{m(2\pi)}{k \sin\theta} = m\frac{\lambda}{\sin\theta}$$

$$kx \sin\theta = (2m+1)\pi \quad \text{Minima}$$

$$x_{min} = \frac{(2m+1)\pi}{k \sin\theta} = \left(\frac{2m+1}{2}\right)\frac{\lambda}{\sin\theta}$$





Applications: Lithography

Example. Calculate the required angle to produce a 1 μm feature using blue light ($\lambda = 450\ nm$).

Interference (3)

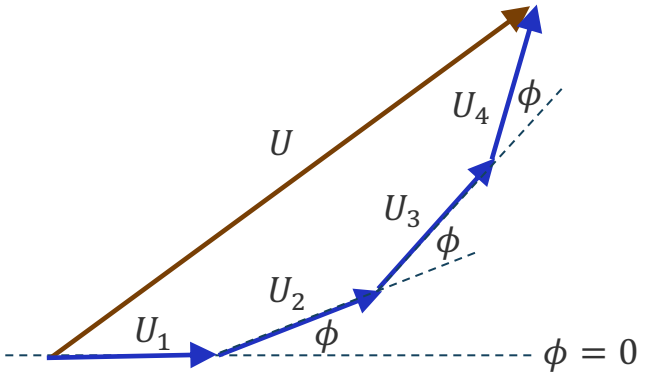
3) Multiple wave interference. Consider M monochromatic plane light waves with same amplitude and same wavelength. Constant phase shift ϕ between each.

$$U_m = \sqrt{I_o} \cdot e^{j(m-1)\phi} \qquad m = 1, 2, \dots, M$$

$$U(r) = \sum_{1}^{M} U_m = \sqrt{I_o}\left(1 + e^{j\phi} + e^{j2\phi} + \cdots + e^{j(M-1)\phi}\right)$$

$$U(r) = \sqrt{I_o}\left(\frac{1 - e^{jM\phi}}{1 - e^{j\phi}}\right)$$

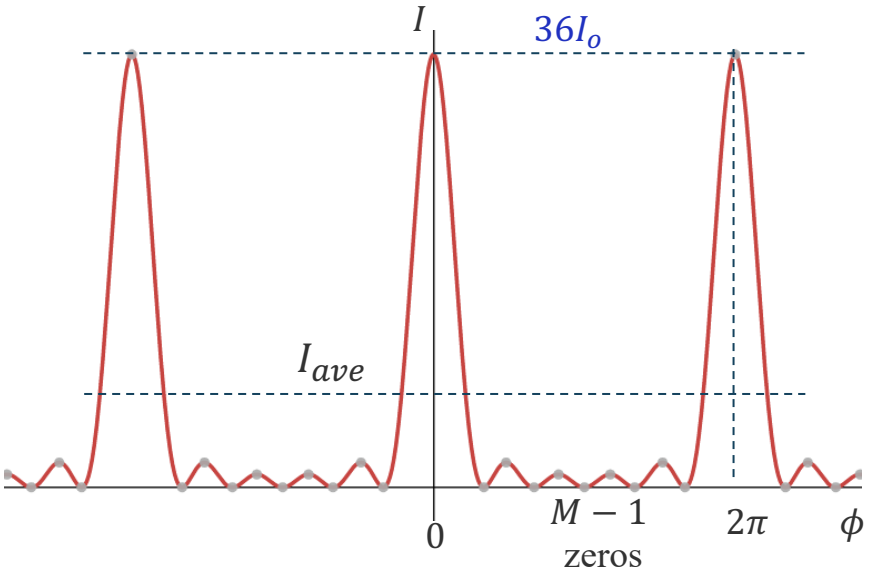
$$I(r) = |U(r)|^2 = I_o \left|\frac{1 - e^{jM\phi}}{1 - e^{j\phi}}\right|^2 = I_o \frac{sin^2(M\phi/2)}{sin^2(\phi/2)}$$



Example. Draw the interference pattern for six sources with I_o. Find the maximum and average intensity.

$$I_{max} = M^2 I_o$$

I_{ave} =? Assignment



Possible application: intense light

Interference (4)

4) Two spherical monochromatic light sources, in-phase with same wavelength.

For spherical waves,

$$U(r) = \frac{A}{r} e^{-jkr}$$

Phase difference is related to

$$r_2 - r_1$$

Constructive interference for

$$r_2 - r_1 = n\lambda \qquad n = 0, \pm 1, \pm 2, \ldots$$

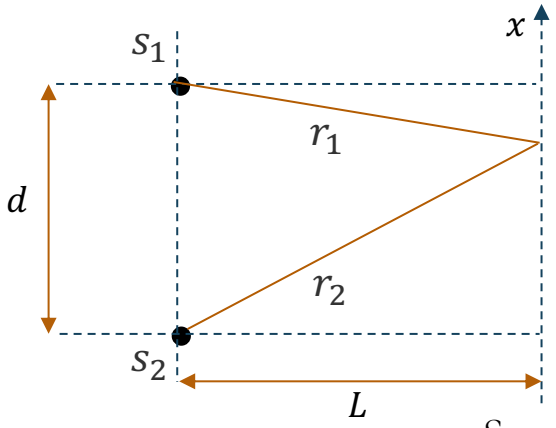
Destructive interference for

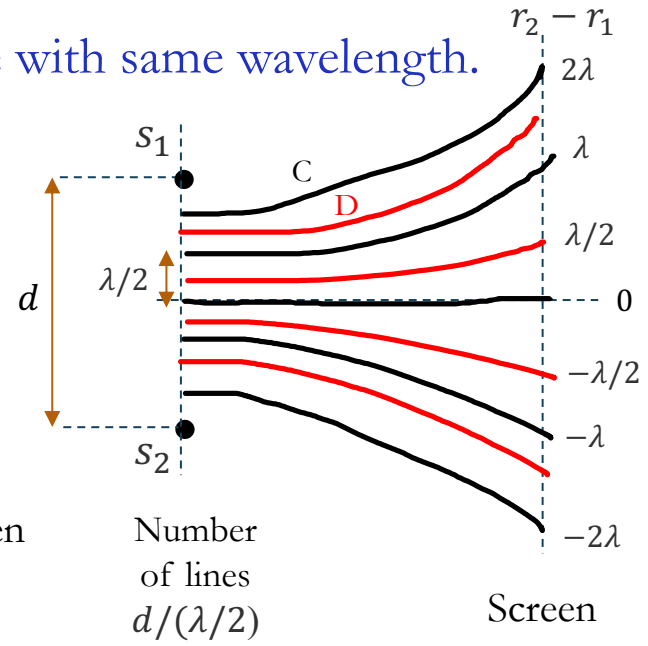
$$r_2 - r_1 = \frac{(2n+1)\lambda}{2}$$

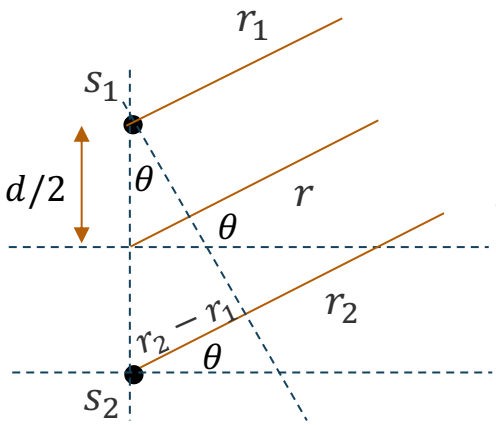
Locations of max/min on the screen (assume $r \gg d$)

Phase difference is given by

$$\delta = \frac{2\pi}{\lambda}(r_2 - r_1) = \frac{2\pi}{\lambda}(d \sin\theta)$$







Constructive interference

$$d \sin\theta = n\lambda \quad \rightarrow \sin\theta_n = n\lambda/d$$

$$x_n = L \tan\theta_n \approx L \sin\theta_n = nL\lambda/d$$

Destructive interference

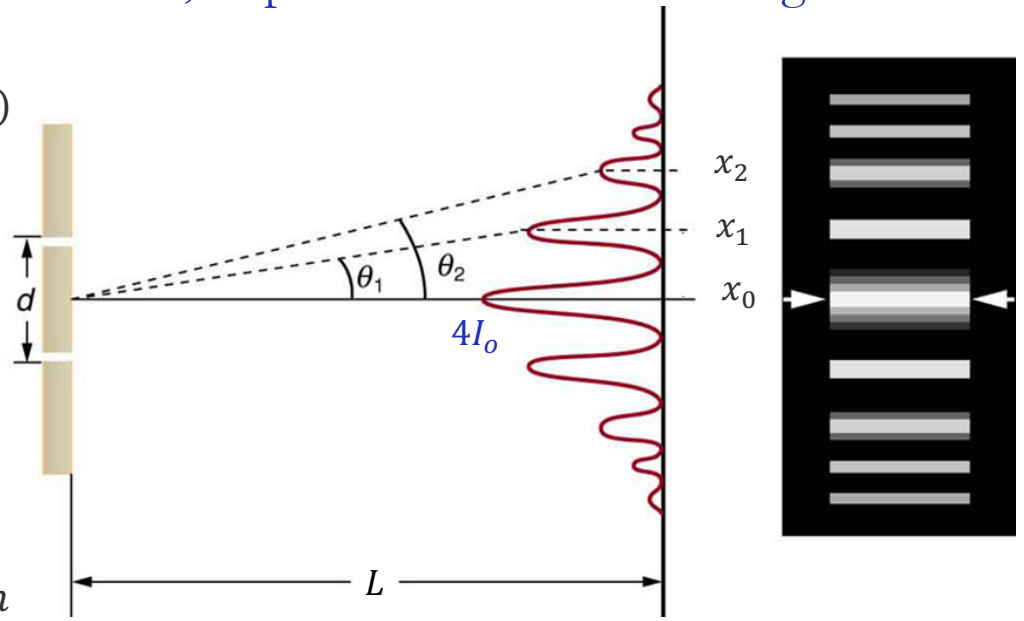
$$x_n = \frac{(2n+1)L\lambda}{2d}$$

Interference (4)

4) Two spherical monochromatic light sources, in-phase with same wavelength.

$$I_n = I_o \left(\frac{\sin \delta_n}{\sin \delta_n/2}\right)^2 \qquad \delta_n = \frac{2\pi}{\lambda}(d \sin \theta_n)$$

$$I_n = 4I_o(\cos \delta_n/2)^2$$



Example.

d=68 μm

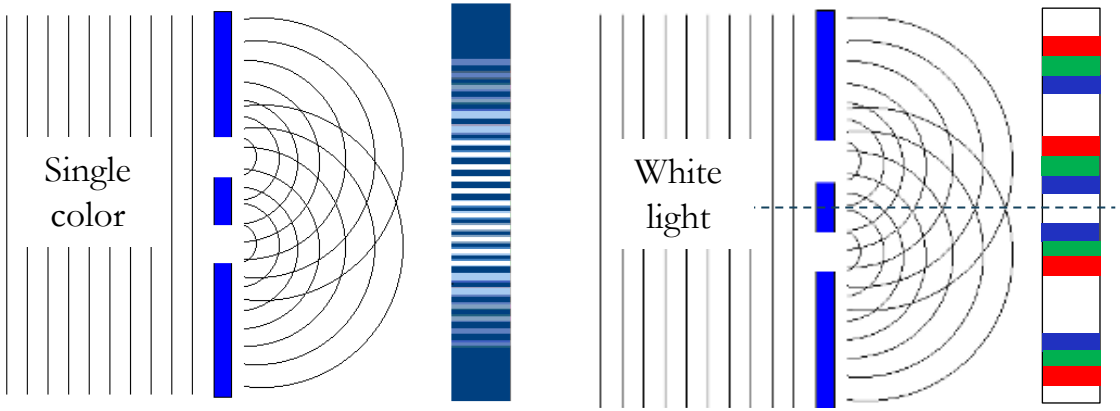
λ=632.8 nm (red)

L=10 m

$\theta_o = 0,\ x_o = 0,\ \theta_1 = 0.412°,\ x_1 = 7.2\ cm$

Question. How are the two sources realized in practice?

Question. If white light is used, how does the interference pattern look like?



Interference (5)

5) Many (N) spherical monochromatic light sources, in-phase with same wavelength. The Gratings problem.

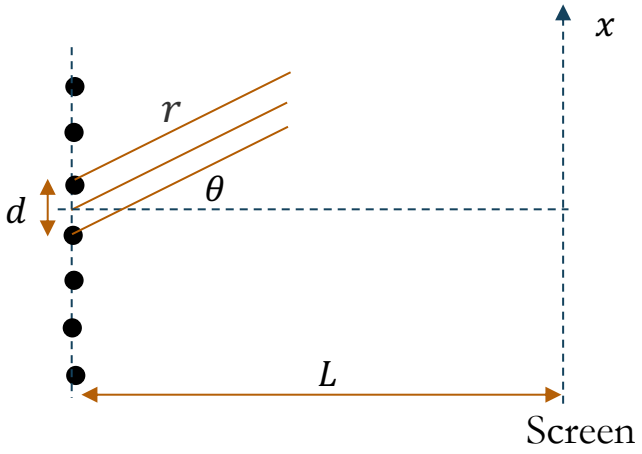
Look at the problem 2 sources at a time.

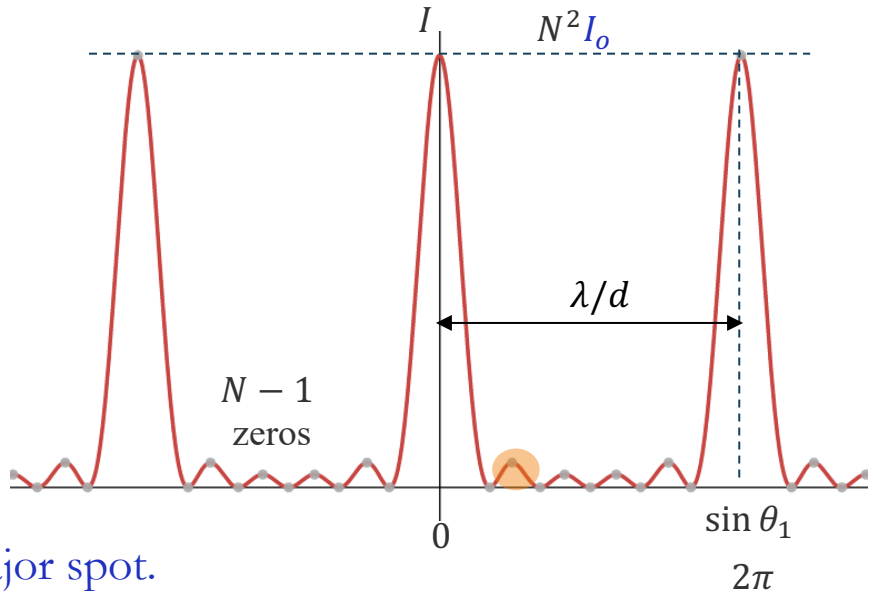
$$\delta = \frac{2\pi}{\lambda}(d \sin\theta_n) = n2\pi$$ Constructive

$$\sin\theta_n = n\lambda/d \qquad x_n \approx nL\lambda/d$$

$$I = I_o\left[\frac{\sin(N\,\delta/2)}{\sin(\delta/2)}\right]^2 \qquad I_{max} = N^2 I_o$$

$$\delta = 0, 2\pi, 4\pi, \ldots$$





Example.
d=2 μm
λ=532 nm (green)
Spot size=2 mm
(1000 lines)
L=5 m → Δx=1.3 m



Find the intensity of the first max after the major spot.

Interference (6)

6) Single slit interference. The Diffraction problem.

Phase difference is related to

$$\beta = \frac{2\pi}{\lambda}(r_2 - r_1) = \frac{2\pi}{\lambda}\frac{a}{2}\sin\theta$$

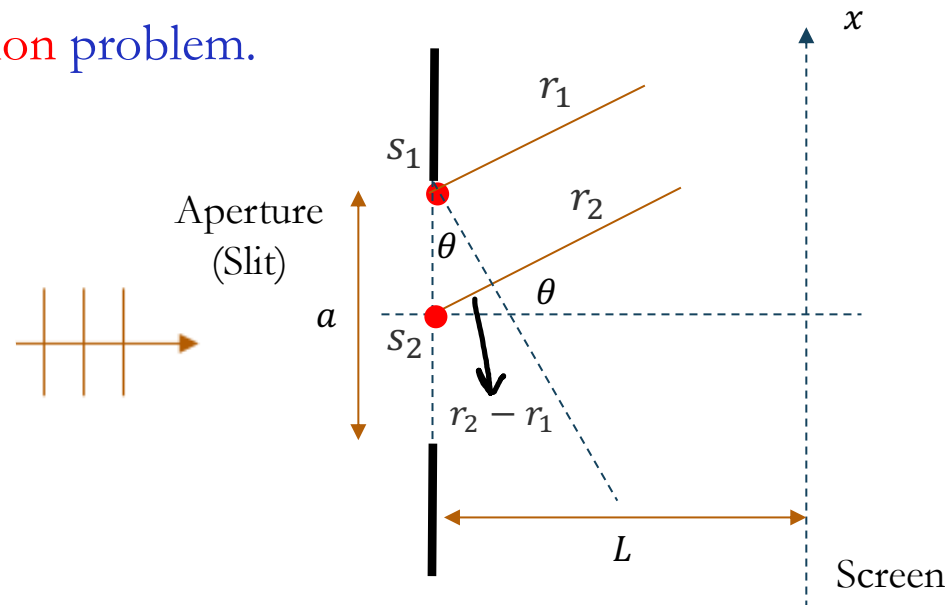
Destructive interference with

$$\frac{a}{2}\sin\theta = n\frac{\lambda}{2} \quad \rightarrow \sin\theta = n\frac{\lambda}{a}$$

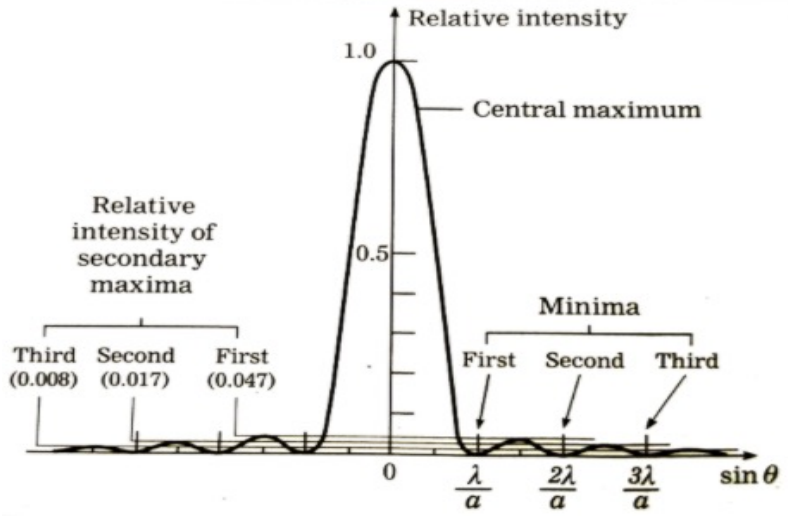
Constructive interference max at $\theta = 0$

Each two points produce the same effect.

Intensity is given by $$I(\beta) \approx I_o\left(\frac{\sin\beta}{\beta}\right)^2$$



$\sin\theta$	β	$\sin\beta$	I	x
0	0	0	I_o	0
λ/a	π	0	o	$\lambda L/a$
$2\lambda/a$	2π	0	o	$2\lambda L/a$



Interference (6)

6) Single slit interference. The Diffraction problem.

Example.1. Find the spot size projected on a screen (L=3 m) by a red laser (630 nm) through an aperture of 0.1 mm.

$$x_1 = \frac{\lambda L}{a} = 2\ cm, x_{-1} = -2\ cm$$ Spot size on screen= 4 cm

Example.2. A 550-nm light through an aperture produces the dark bright spot at 45°. Find the aperture size.

$$a \sin\theta_2 = 2\lambda \qquad a = \frac{2(550\ n)}{\sin 45} = 1.56\ \mu m \qquad \theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 20.7°$$

Example.3. A 589-nm light goes through a 7.5-μm aperture find the first few angles of the dark spots.

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 4.5° \qquad \theta_2 = 9.05° \qquad \theta_{13} = NA$$

Example.4. A 550-nm light goes through a 1-μm aperture find the first few angles of the dark spots.

$$\theta_1 = \sin^{-1}\left(\frac{\lambda}{a}\right) = 33.4° \qquad \theta_2 = NA$$

Interference (6)

6) Single slit interference. The Diffraction problem.

The light intensity of a grating becomes

Some bright spots in the gratings pattern can possibly be nulled by dark spots in the diffraction pattern.

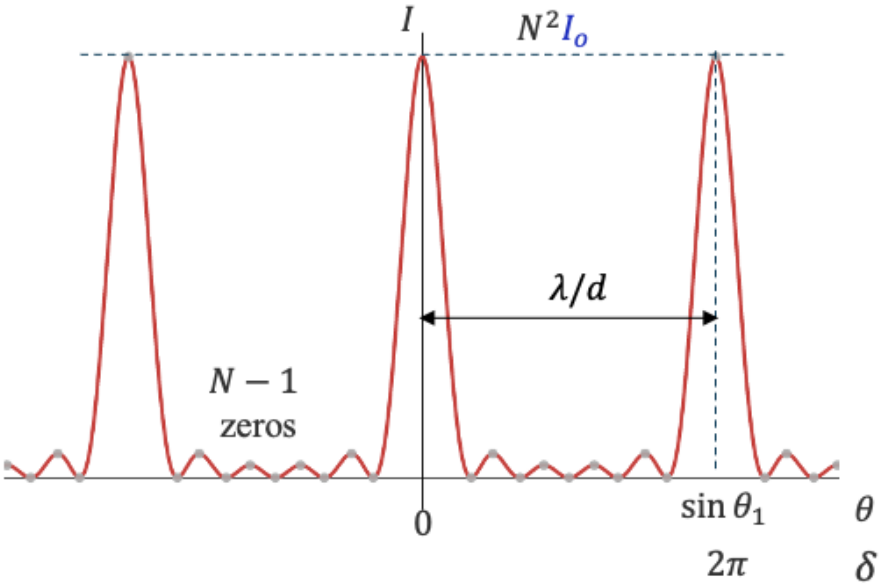
$$I = I_o \left[\frac{\sin(N\,\delta/2)}{\sin(\delta/2)}\right]^2 \cdot \left(\frac{\sin\beta}{\beta}\right)^2$$

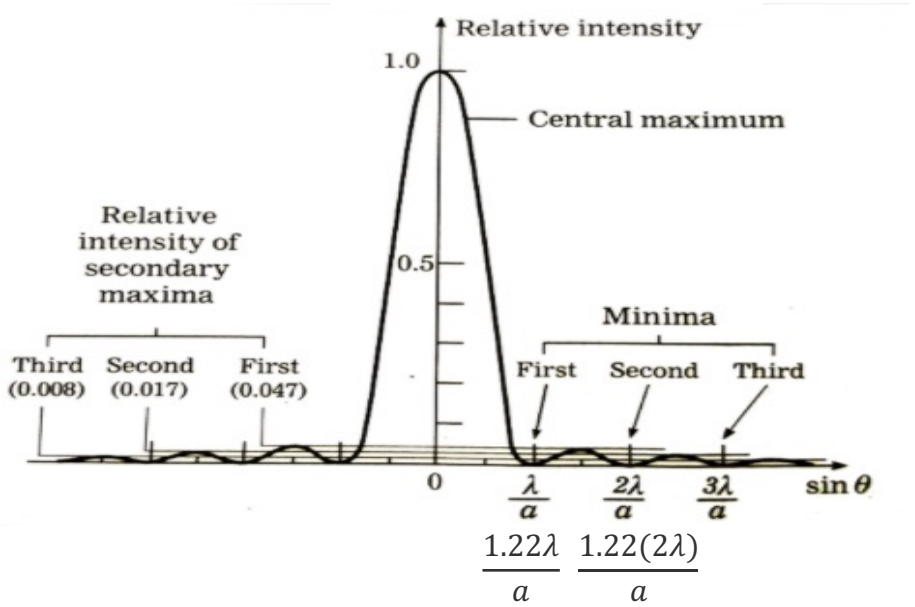
Interference Diffraction

$$\delta = \frac{2\pi}{\lambda}(d\sin\theta)$$

$$\beta = \frac{\pi a}{\lambda}(\sin\theta)$$

For example, if $a = d/5$, the 5th order bright spot in the grating pattern is lost because of the null in the diffraction pattern.

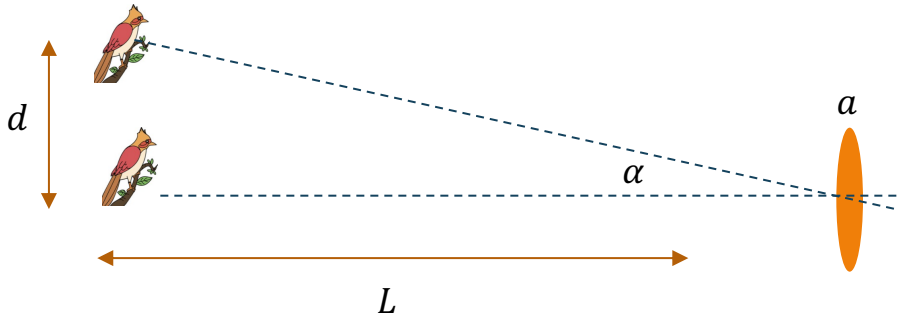


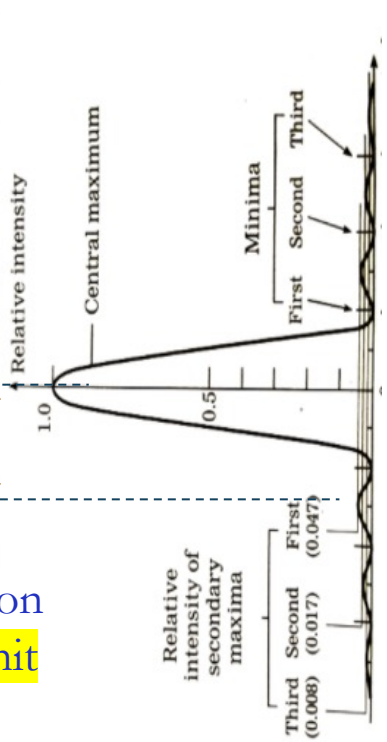


Interference (6)

6) Single slit interference. The Diffraction problem.

For a circular aperture (e.g., human eye) $\frac{\lambda L}{a} \rightarrow 1.22\,\frac{\lambda L}{a}$





$\tan\alpha = \frac{d}{L} \rightarrow \alpha = \tan^{-1}\frac{d}{L}$ $\quad \alpha \geq \Delta\theta$

Resolution condition

The diffraction limit

For the human eye, a=4 mm, λ=500 nm $\quad \Delta\theta = 1.5\times10^{-4}$ rad

For Hubble telescope, a=2.4 m, λ=500 nm, $\Delta\theta = 0.28\times10^{-6}$ rad

Q & A

Send me an email if you have any questions.